

Material Response Under Mechanical Load

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Abstract—The test's objective is to understand material responses (brittle and ductile) under mechanical loading using a Universal Testing Machine (UTM), learn techniques to measure stress and strain, and identify various material properties from a simple uniaxial test.

I. INTRODUCTION

A. Defining strength, stiffness, toughness, ductility, hardness, brittleness, flexure strength, and strain energy, considering the material response.

- **Strength** refers to the ability of a material to withstand a force without breaking or deforming permanently.
- **Stiffness** of a material is the measure of its ability to return to its original form after being loaded by an external force.
- **Toughness** of a material is defined as the ability of a material to absorb energy and plastically deform before fracture takes place.
- **Ductility** measures a material's ability to deform plastically (permanent deformation) before fracture.
- **Hardness** can be defined as the ability of a material to withstand force without deformation, penetration and indentation.
- **Brittleness** of a material is defined as the nature that when a material is loaded with a certain amount of force, it breaks or fails suddenly with negligible plastic deformation.
- **Flexural strength** is defined as the material's ability to resist deformation under load, regardless of the direction of loading.
- **Strain energy** is defined as the energy stored within a material or body when it undergoes deformation due to applied forces.

B. What is the working principle of a Load Cell, and what are the types of load cells?

- A load cell is a transducer which converts force into a measurable electric output. The most common load cells are the strain gauge, which offer accuracy within a range of 0.03% to 0.25%.
- The working principle of load cells is as follows:- When an applied load causes deformation, the resistance of the strain gauge (load cell) alters, which is measured by a Wheatstone Bridge configuration. The electric output (Voltage, current) changes accordingly and is proportional to the applied load.
- Strain Gauge Load Cells, Hydraulic Load Cells, Pneumatic Load Cells, Capacitive Load Cells.

C. How can we measure the strain? What types of devices are used for measuring strains?

- We can measure strain by converting the elongation or compression into a meaningful electrical signal, which can then be mapped to give us the deformation.
- Strain can be measured using multiple devices; some of the most commonly used are extensometers, which we used in the experiment to measure the change in the length of the specimen, and another widely used device is a strain gauge.

D. Why are there pre-defined geometries (ASTM standards) for testing out materials under compression (typically cylindrical) and tension (dog-bone samples)?

- The Young's Modulus, Yield Stress, and Ultimate Strength are intrinsic properties of a material. They are geometry-independent but the standardisation of dimensions and testing procedures is crucial for ensuring consistency, comparability, and reliability of the test results across different laboratories and industries.
- The specimen's shape during a tensile test is chosen as a dog bone to ensure that stress concentration in the narrow section is higher, ensuring that the material will break in the gauge section.
- The shape of the specimen during a compressive test is chosen as cylindrical to ensure uniform stress distribution in the specimen, and as the shape is uniform, it ensures that the specimen does not buckle in a particular direction during the test.

II. PROCEDURE

- 1) Mark the gauge length in each of the specimens and measure the dimensions (width, thickness, length) of the specimens using a Vernier caliper.
- 2) Select an appropriate UTM and load cell for the experiment and check if the connected digital device(controller + data acquisition) is working properly.
- 3) Erase all previously collected data from the device.
- 4) Carefully clamp the specimen between the top and bottom grips of the UTM.
- 5) Clamp an extensometer to the specimen with the help of rubber bands positioning the edges slightly outside the gauge markings on the specimen.
- 6) Connect the extensometer cable to the digital device of UTM.
- 7) Measure the grip length using a vernier calliper.

- 8) Fill in the required data to the UTM digital device like width, thickness, gauge length, grip length.
- 9) Set the cross-head speed of the UTM to 5 mm/min.
- 10) Run the experiment by clicking the start button on the device.
- 11) The test stops automatically and the timer stops upon the failure of the specimen.
- 12) Carefully clamp out the extensometer and the specimen from the grips.
- 13) Save the data and use a pendrive to transfer the collected data to the computer for analysis.
- 14) Repeat the procedure for the other specimens.

III. OBSERVATIONS

We obtained the Load (in kg) vs Extension (in mm) from the Universal Testing Machine (UTM) by using extensometer (as UTM machine was unable to provide it) for three materials (and three trials for each): (i) Aluminum, (ii) Mild Steel and (iii) Polymer.

The data recorded (for each trial) has multiple data points, hence the **link of the files of raw data and also calculated stress-strain is attached at the end**. Here, we only show the resulting stress - strain curve of each trail.

A. Calculating Stress and Strain:

The (axial) stress (σ) on the specimen can be calculated as:

$$\sigma = \frac{Pg}{A_c} = \frac{Pg}{tw}$$

where, $P(kg)$ is the load measured by UTM, $g(m/s^2)$ is the acceleration due to gravity, $t(mm)$ is the thickness and $w(mm)$ is the gauge width of the specimen.

The (axial) strain (ϵ) of the specimen is calculated as:

$$\epsilon = \frac{\Delta L}{L}$$

where, $\Delta L(mm)$ is the extension given by extensometer and $L(mm)$ is the initial length of the gauge section.

All had gauge length $L = 36$ mm and thickness $t = 3$ mm. The width w of gauge of aluminum, mild steel and polymer respectively are 6.4, 5.5 and 6 mm. The graphs of each trial for each material and fractured specimen is tabulated below.

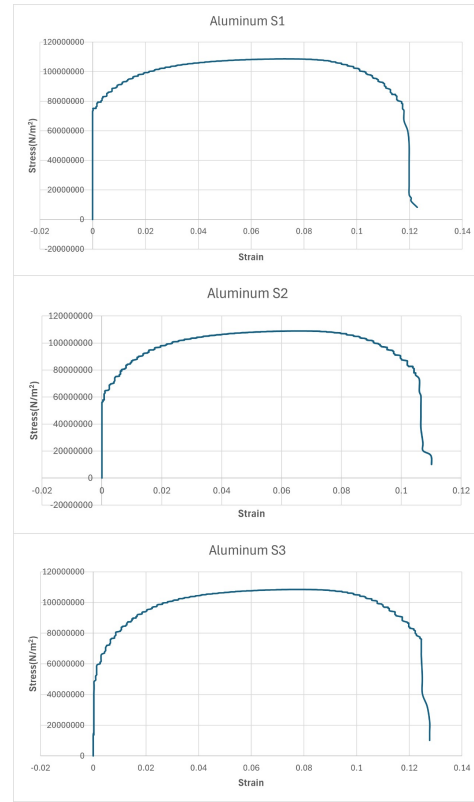


Fig. 1: Stress-strain plots of each trial for Aluminum.

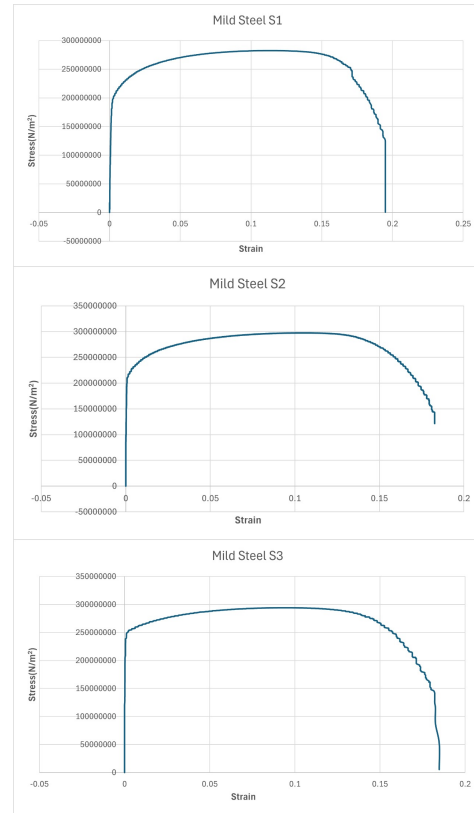


Fig. 2: Stress-strain plots of each trial for Mild Steel.

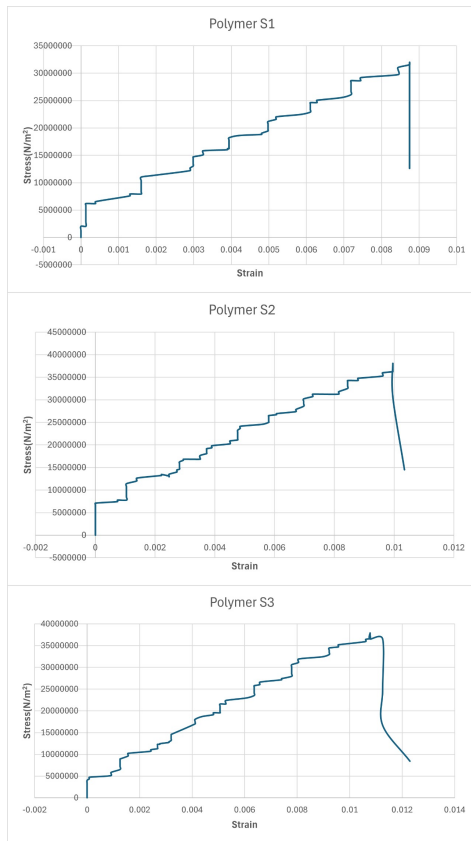


Fig. 3: Stress-strain plots of each trial for Polymer.



Fig. 4: Fractured specimens of Aluminum, Mild Steel, and Polymer respectively.

From data of three trials for each material, we determined the average Ultimate Tensile Strength (UTS), Fracture Strength, Strain at Fracture and Time Taken for the specimen to fracture.

Aluminum: UTS = 108.63 MPa, Fracture Strength = 73.69 MPa, Strain at Fracture = 0.1161, Time = 109 s.

Mild Steel: UTS = 291.15 MPa, Fracture Strength = 124.74 MPa, Strain at Fracture = 0.1875, Time = 310 s.

Polymer: UTS = 35.97 MPa, Fracture Strength = 33.01 MPa, Strain at Fracture = 0.009985, Time = 16 s.

The analysis and interpretation of the values with actual specimen behavior is given below.

IV. INTERPRETATION AND COMPARISON

- 1) First, We observe that both Aluminum and Mild Steel have much higher strength (UTS) and strain (before

fracture) as compared to polymer whose both strength and strain are much smaller (Figures 1, 2 & 3). Therefore, the polymer shows brittle behavior as expected and aluminum & mild steel are more ductile.

- 2) Another difference that is observed between these ductile and brittle material is the time taken for fracture. Brittle materials fracture faster than ductile under tensile loading and hence absorb less strain energy than ductile ones.
- 3) Finally, ductile materials have significant difference in their UTS and fracture strength indicating neck formation (high deformation) usually beyond UTS till fracture as seen in above images (First two of Figure 4) of fractured aluminum and mild steel specimens. Whereas brittle materials do not have this huge difference and break shortly after UTS with no deformation and clean - plane break as seen for fractured polymer specimen (Last of Figure 4).
- 4) Among aluminum and mild steel, mild steel showed more strength (UTS), strain, time taken for fracture and hence strain energy stored than aluminum implying that mild steel behaved more ductile than aluminum in this case.

V. CONCLUSION

The experimental results clearly demonstrate that ductile materials such as mild steel and aluminum absorb significantly more strain energy than brittle materials like polymer. Ductile materials exhibited gradual failure, showing necking before fracture, whereas brittle materials fractured suddenly with negligible deformation. Furthermore, brittle specimens required much less time to fail under the same strain rate compared to ductile ones. Among the ductile materials tested, mild steel showed higher strength, greater strain at fracture, and longer time to failure than aluminum, confirming its superior ductility and toughness.

DATA LINK

The link for the raw data and calculation files (Excel):

<https://github.com/MukeshK17/>

Material-Response-Under-Mechanical-Load.git

REFERENCES

- [1] RapidDirect, "Stiffness of Materials," [Online]. Available: <https://www.rapiddirect.com/blog/stiffness-of-material/>
- [2] Fictiv, "Engineering Fundamentals: Ductility, Strain, and Toughness," [Online]. Available: <https://www.fictiv.com/articles/engineering-fundamentals-refresh-ductility-strain-and-toughness>
- [3] RapidDirect, "Hardness of Materials," [Online]. Available: <https://www.rapiddirect.com/blog/hardness-of-materials/>
- [4] ScienceDirect, "Flexural Strength," [Online]. Available: <https://www.sciencedirect.com/topics/earth-and-planetary-sciences/flexural-strength>
- [5] ScienceDirect, "Brittleness," [Online]. Available: <https://www.sciencedirect.com/topics/materials-science/brittleness>
- [6] Tacuna Systems, "Overview of Load Cells," [Online]. Available: <https://tacunasystems.com/knowledge-base/an-overview-of-load-cells/>
- [7] ResearchGate, "Why there are ASTM Standards for Tensile Testing Dimensions," [Online]. Available: https://www.researchgate.net/post/Why_there_is_astm_standards_for_tensile_testingdimensions